



PT.SUMITOMO ELECTRIC WINTEC INDONESIA

No. Dokumen

4-QAS-PP07-001

Revisi

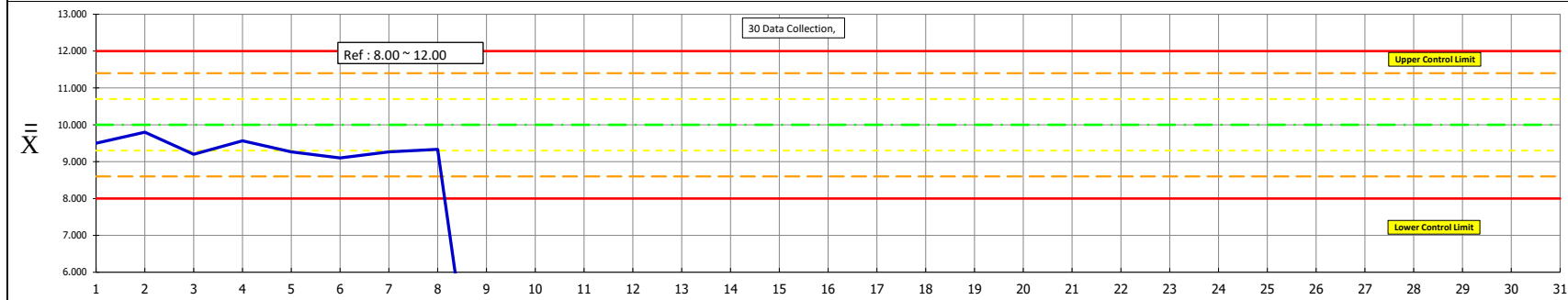
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Tgl Berlaku

2-Dec-16

 $\bar{X}$ - R CONTROL CHART

MACHINE DF Tester (Old)	SECTION QC	OPERATION DF 280° C	DATE CONTROL LIMITS CALC.	SPECIFICATION min 2 max 20	COURSE NO.	CONTROL ITEM (Δ) YES ☐ NO ☐
	MONTH May-25	CHARACTERISTIC Temperatur	SAMPLE SIZE / FREQUENCY n = 3	TYPE / SIZE : D2IDW4-B-2 / 0.70	PIC : DWI AS	

 $\bar{X}$ = AVERAGE $\bar{X}$ = **10.000**U.C.L. = $\bar{X} + A_2 R$ = **12.0000**L.C.L. = $\bar{X} - A_2 R$ = **8.0000**AVERAGE ($\bar{X}$ BAR CHART)

SPECIAL CAUSES

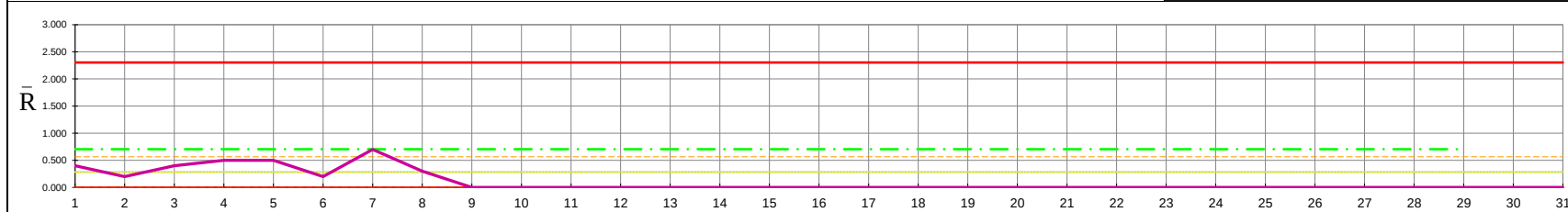
- ☒ TEST 1: ANY POINT OUT OF CONTROL LIMIT
- ☒ TEST 2: 7 POINTS IN A ROW ON SAME SIDE OF CENTER LINE
- ☒ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
- ☒ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
- ☒ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
- ☒ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
- ☒ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
- ☒ TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

ACTION INSTRUCTION

- _____
- _____
- _____
- _____
- _____

 $\bar{R}$ = AVERAGE $\bar{R}$ = **0.705**U.C.L. = $D_4 \bar{R}$ = **2.303**L.C.L. = $D_3 \bar{R}$ = **0.00**RANGES ($\bar{R}$ - CHART)

Special Cause Marker Legend:

Test 1 ● Test 2 ◆ Test 3 ▲ Test 4 ■

SUBGROUP SIZE	A ₂	d ₂	D ₃	D ₄
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.135
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

Special Cause	Titik berada di luar garis merah atau lebih dari 3 σ	6 titik berada dalam satu urutan secara terus menerus, baik posisi naik ataupun turun	4 atau 5 titik lebih dari 1 σ dengan posisi yang sama dan titik keluar dari garis merah atau 3 σ
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Month		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Date																																
DATA	1	9,500	9,700	9,200	9,600	9,300	9,100	9,300	9,500																							
	2	9,300	9,800	9,400	9,300	9,500	9,000	9,600	9,300																							
	3	9,700	9,900	9,000	9,800	9,000	9,200	8,900	9,200																							
N		3	3	3	3	3	3	3	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AVERAGE, X		9,500	9,800	9,200	9,567	9,267	9,100	9,267	9,333	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Range		0,400	0,200	0,400	0,500	0,500	0,200	0,700	0,300																							
Moving Range			0,002	0,768	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,000
SHIFT 1			dwas	dwas	dwas			dwas																								
SHIFT 2																																

CALCULATION

$$\begin{aligned} \bar{X} &= 9.3792 \\ R &= 0.4000 \\ S_{ST} &= 0.2363 \quad S_{ST} = 0.283 \\ C_{PK} &= \min \left[\frac{\bar{X} - LSL}{3S_{ST}}, \frac{USL - \bar{X}}{3S_{ST}} \right] \\ &= \min \left[\frac{1.95}{3.70}, \frac{3.70}{1.95} \right] \\ &= 1.95 \\ Cp &= \frac{USL - LSL}{6S_{ST}} \\ &= \frac{2.82}{6 \times 0.2363} \end{aligned}$$

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